Capstone Certificate Track Proposal

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Certificate Titles: IBM AI Engineering Professional Certificate | Generative AI with Large Language Models | Google Project Management Professional Certificate

1. Professional Interests and Career Direction

What first got me into AI/ML really more in depth was after I took my first Data Science class and bridged that with AI to understand my foundation. After that, I became very fascinated with AI and its capabilities, how useful and powerful it can be when used correctly. After CST 383 I started working on a computer vision project with a graduate student and Dr. Feng, which involved a lot of ML topics as well as foundational skills needed for AI. Working on this Basketball Action Detector was my first time using Hugging Face and accessing different trained models, and learning about training my own models for certain tasks.

I don't have a strong pull towards one specific industry. I'm great working in a small or large company or team. What matters to me is being in an environment where I can be productive, learn, and apply skills to real world situations. I'm driven to want to create something that can be useful for others, everyday things that can be automated or made more accessible and simple.

Near term, my plan is to finish my certificates, keep building on my interests like the B.A.D. project, and keep applying for entry-level AI/ML engineering roles as I get closer to graduation. In the meantime I've been doing freelance web design work. I've built websites for small businesses these past six months, which has taught me some useful frontend design skills along the way.

2. Gap Analysis: Knowledge and Skill Gaps

A lot of AI roles want work experience already, or prefer people who've been doing ML for years already. There's always more to learn, and I feel like I have the right tools to learn what these jobs have to offer. I looked at postings for entry-level and intern AI/ML roles, including the Oklahoma City Thunder's Applied AI Engineer Intern and Software Engineer Intern postings (Basketball Operations), an Intel Software Engineering Intern posting, Atlassian's 2026 grad ML Engineer posting, and an entry-level Data Scientist role at CoLab in Portland. Cross-checking those against a 2026 analysis of roughly 3,449 AI job postings on the skills employers ask for most (Python, LangChain, cloud AI deployment, LLM integration) here's where my background has real gaps.

- **Skill/Knowledge Gap 1 – Production LLM and agent tooling (RAG, LangChain, prompt engineering at a systems level).** My coursework and B.A.D. work involve calling and evaluating LLMs like Qwen3-VL, but I haven't built or deployed a RAG pipeline or agent system end to end. The Thunder's Applied AI Engineer Intern posting explicitly lists experience building RAG pipelines and fine-tuning foundation models as

a qualification, and the wider market analysis flags LangChain and LLM integration as the top-cited skills in entry-level AI engineer listings.

- **Skill/Knowledge Gap 2 – Cloud deployment and MLOps.** B.A.D. runs out of Colab notebooks, which is fine for research but isn't how production ML gets shipped. The same Thunder posting asks for experience with cloud-ML tooling like AWS SageMaker and Bedrock, and advanced SQL against warehouses like Redshift and Athena, which I don't have hands-on yet.
- **Skill/Knowledge Gap 3 – Still building depth on the fundamentals side (classical ML and project/delivery management).** I feel decent here since I'm actively working on both through the certs I picked, but there's always more depth to get. My background is basketball training and commission sales, not structured project delivery, so learning to manage my own workstream (planning, stakeholder communication, delivery timelines) the way these postings expect is something I'm still building.

Postings reviewed:

- Applied AI Engineer Intern / Basketball Operations – Oklahoma City Thunder
- Software Engineer Intern / Basketball Operations – Oklahoma City Thunder
- Software Engineering Intern, Bachelor's – Intel Corporation
- Machine Learning Engineer, 2026 Graduate – Atlassian
- Data Scientist (entry level) – CoLab, Portland, OR

3. Certificates of Your Choice

Hours below are the estimates from the course-tracking spreadsheet. I dropped Google Advanced Data Analytics from my original list since it's optional and doesn't map as tightly to the applied ML/AI direction, and its hours alone (120-180) would push the total even higher. Even with just these three certs, the combined range is 275-400 hours, past both the 160-200 target in these instructions and the rubric's 140-160 range. I looked at cutting Google Project Management to land closer to that target, but decided to keep all three, since PM reinforces skills I want anyway (structured delivery, working through projects as a team), even if it means going over on hours.

Certificate	Provider	Sign-Up Link	Est. Hours	Status
IBM AI Engineering Professional Certificate	IBM / Coursera	coursera.org/professional-certificates/ai-engineer	140-180	In progress (course 1 of 13 complete, course 2 in progress)
Generative AI with Large Language Models	Coursera	coursera.org/learn/generative-ai-with-llms	15-40	Purchased, not started
Google Project Management (Career	Google / Coursera	grow.google/certificates/project-	120-180	In progress (course 1 of 7 complete, course

Certificate	Provider	Sign-Up Link	Est. Hours	Status
Certificate)		management/		2 in progress)

Planned Timeline

Weeks	Focus	Hours
1-6	Finish IBM AI Engineering courses 2-6 (Deep Learning with Keras/TensorFlow/PyTorch, course 6 capstone)	~70-90
7-8	Generative AI with Large Language Models (full course)	~15-40
9-14	IBM AI Engineering courses 7-13 (Gen AI, LangChain, RAG, agents, course 13 project)	~70-90
Ongoing	Google Project Management, self-paced alongside the above, likely finishing after the Dec 2026 grad date	~120-180

4. How the Certificates Bridge Your Skill Gaps

IBM AI Engineering is the big one, 13 courses covering a lot of foundation, starting with classical ML in Python (course 1, which I've completed) and working through deep learning with Keras, TensorFlow, and PyTorch. Course 6 is its own capstone project, and the back half is where it hits Gaps 1 and 2 directly: courses 12 and 13 are "Fundamentals of AI Agents Using RAG and LangChain" and a hands-on RAG/LangChain project, exactly the production LLM/agent tooling I'm missing right now.

Generative AI with Large Language Models is more focused. It's helped a lot with going through fine-tuning and getting better accuracy and calculations out of a model, which is directly useful for Gap 1 since it's teaching me how to actually make an LLM perform well for a specific task instead of just calling one.

Google Project Management reinforces the ideals of being a software engineer: keeping code a specific way, thinking about scalability, working through projects efficiently as a team, and handling faults that come up in any work environment. That maps onto Gap 3, since it's the structured-delivery side of things I don't get much of from CS coursework alone.

5. Applied Projects and Deliverables

Each certificate comes with its own built-in project I can use as the deliverable for this class. IBM AI Engineering has two: the "AI Capstone Project with Deep Learning" (course 6 of 13) and the "Project: Generative AI Applications with RAG and LangChain" (course 13 of 13), which is a hands-on build using the RAG/LangChain skills tied to Gap 1. Google Project Management has

its own capstone too, “Capstone: Applying Project Management in the Real World” (course 6 of 7), applying the full project lifecycle to a real scenario. Generative AI with Large Language Models doesn’t come with a built-in capstone project the same way, so I’ll build my own project applying what I learn from that course (fine-tuning and improving accuracy/output from a model) once I get into it.

Timeline: IBM’s course 6 capstone first, then Google PM’s course 6 capstone once I get there, then IBM’s course 13 RAG/LangChain project, then my own Gen AI with LLMs project last, running from now through graduation in December 2026.

Portfolio presentation: I have a portfolio site (ryebriscoe.github.io) and GitHub (github.com/ryebriscoe). I’ll add write-ups for each certificate project as I finish them, plus screenshots/demos and my certificate completions linked from the portfolio site.